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# Fountain View Hall

*Release 0.2.0*

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Fountain View Hall's management tool, `fvh-manager` is used to manage reservations at Fountain View Hall.



## USER GUIDE

This guide will walk you through `fvh-manager` and covers topics such as how to install and use `fvh-manager`.

### 1.1 Installation

#### 1.1.1 User Prerequisites

To verify you meet all prerequisites, run:

```
uv --version
```

Expected output:

```
uv 0.x.x
```

Not:

```
command not found: uv
```

#### Important

Make sure the `uv` command runs before running the program.

If `uv --version` fails, use [this guide](#) to install `uv`.

#### 1.1.2 Minimal Getting Started

Make sure you run these commands from the project directory

Run this once:

```
uv sync
```

Then run this to start on every subsequent run:

```
uv run fvh-manager
```

### 1.1.3 Activate Completion by Operating System and Shell

If desired, you can activate shell completions by running one of the following commands, depending on your operating system and shell:

| Operating system | Shell      | Activation command                               |
|------------------|------------|--------------------------------------------------|
| Linux/macOS      | Bash       | <code>source completions/fvh-manager.bash</code> |
| Linux/macOS      | Zsh        | <code>source completions/_fvh-manager</code>     |
| Linux/macOS      | Fish       | <code>source completions/fvh-manager.fish</code> |
| Windows          | PowerShell | <code>. .\completions\fvh-manager.ps1</code>     |

Once shell completions are activated, make sure your virtual environment is active. You can then run the program with:

```
fvh-manager
```

To view available completions, type:

```
fvh-manager <tab>
```

## 1.2 Developer Installation

Meet all previous *prerequisites* and additional ones as specified bellow.

Install developer dependencies for python:

```
uv sync --all-groups
```

Verify that `pdflatex` is installed:

```
pdflatex -v
```

Verify that `git` is installed:

```
git -v
```

Verify that `lefthook` is installed:

```
lefthook -v
```



## REFERENCES

Command-line entry point for the project.

**author**

Derek R. Neilson

This module defines the project's Typer application and should remain lightweight by design.

Typer is a command-line interface (CLI) framework used throughout this project. It provides decorators such as `typer.command()` for registering CLI commands.

 **See also**

[Typer documentation](#)

`src.main.create_engine(in_memory)`

Create a SQLite database engine.

Creates either an in-memory SQLite database or, by default, a persistent database at `DB_PATH`, which points to `database/data.db`.

**Parameters**

**`in_memory`** (bool) – If `True`, create an in-memory database. If `False`, create a persistent database at `DB_PATH`.

**Returns**

A SQLAlchemy engine representing the configured database connection.

**Return type**

Engine

The SQLAlchemy database URL has the general form:

```
dialect[+driver]://user:password@host/dbname[?key=value...]
```

The default project database URL has the form:

```
sqlite+pysqlite:///database/data.db
```

If `in_memory` is `True`, the database URL has the form:

```
sqlite+pysqlite:///memory:
```

 **See also**[SQLAlchemy Engine documentation](#) [Typer documentation](#)

```
src.main.start_app(in_memory=False)
```

Start the CLI app.

**Parameters**

**in\_memory** (bool) – If True, use an in-memory SQLite database.

**Return type**

None

Sets up logging and metrics for the application.

**author**

Derek R. Neilson

**var logger**

Application logger configured for OpenTelemetry.

**var meter**

OpenTelemetry meter used to create counters and histograms.

This module configures OpenTelemetry logging and metrics, then exposes the `track_command` decorator for tracking CLI command execution.

```
src.observability.configure_logging()
```

Configure and return the application logger.

**Return type**

Logger

```
src.observability.configure_metrics()
```

Configure and return the application metrics meter.

**Return type**

Meter

```
src.observability.track_command(command_name)
```

Track command execution with logs and metrics.

The wrapped function records the number of command runs, active operations, errors, and command duration.

**Parameters**

**command\_name** (str) – Name of the command being tracked. Format not required but should be name-of-command

**Return type**

Callable[[TypeVar(F, bound= Callable[... , Any])], TypeVar(F, bound= Callable[... , Any])]

**Example**

```
from src.observability import logger, track_command

@track_command("some-command")
def some_command():
    logger.info("Command executed")
```

The design process has been documented quite thoroughly, but not all decisions are documented.

## 3.1 Fountain View Hall Pseudocode and User Journeys

### 3.1.1 Problem Statement

The project must meet all *requirements*. Besides those requirements, I have set some personal goals as well:

1. Achieve 100% test coverage.

I have done expensive research<sup>1</sup> on this and have concluded that it does not guarantee code standards, but it also does not hurt code quality and is probably to make you think more about your code.

#### Limitation of Testing

“Program testing can be used to show the presence of bugs, but never to show their absence.”

– Edsger W. Dijkstra [Dijkstra1970]

2. Learn more about the LGTM stack for observability

This does not directly contribute to the end goal but is a skill I would like to learn and am taking the opportunity to do so.

3. Document most things very closely

## 3.2 Overview for Fountain View Hall

### 3.2.1 Overview

Fountain View Hall wants a program that tracks event attendance and projected revenue for multiple bookings.

#### 3.2.1.1 Program Requirements

- Repeatedly prompt for event name, guest count, and estimated revenue
- Continue until the user enters “done”
- Track total events processed
- Track total guests
- Track total projected revenue

---

<sup>1</sup> See *Software Testing and Code Coverage* for a list of sources viewed to come to this conclusion.

- Calculate the average guests per event
- Determine the largest event

### 3.2.1.2 Business Rules

- Pricing Rules
    - Large events receive a discount
  - Operational Rules
    - Every booking must include a guest count
    - Booking total must be calculated before the discount is applied
    - Revenue totals should be tracked
- 

## 3.2.2 Planned Technology Usage

I intended to use various technologies throughout this project; some are very necessary, and others save time. Each technology will be categorized and described below:

### 3.2.2.1 Editing/Developer Tools

- *Git* and *GitHub* as a VCS (version control system).<sup>1</sup>
- *uv* for package, project, and environment management.
- *lefthook* as a Git hook manager.
- *pre-commit* to run pre-commit scripts.<sup>2</sup>
  - *ruff* for Python linting and formatting running as a pre-commit script.
- *Sphinx* to help build this beautiful documentation.
- *pytest* as a testing suite.
- *Open Telemetry Protocol* for observability and log management.<sup>4</sup>

### 3.2.2.2 Python Packages

- *typer* for command line interface completions and argument parsing.
  - *SQLite3* as a database<sup>3</sup>
  - *SQLAlchemy* for object-relational mapping (ORM)
- 

## 3.3 Sources

A curated source list for reStructuredText, OpenTelemetry, SQLAlchemy, and Python tooling.

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<sup>1</sup> Go to the [project repo](#) to see commit history.

<sup>2</sup> All scripts (in the scripts directory) are written for Unix systems and use the bash language.

<sup>4</sup> You can visit their [web page](#) for more details, but essentially it is a web interface that helps you view logs and trends in data. Probably unnecessary, but I was eager to learn how to use this.

<sup>3</sup> See SQLite's [home page](#) for more details.

### 3.3.1 Quick Reference

### 3.3.2 reStructuredText and Sphinx

| # | Source                                                | Creator                                             | Date        | Link                        |
|---|-------------------------------------------------------|-----------------------------------------------------|-------------|-----------------------------|
| 1 | <b>reStructuredText Markup Specification</b>          | Goodger, David                                      | 2026        | <a href="#">Open source</a> |
| 2 | <b>Restructuredtext Primer - Material for Sphinx</b>  | Sheppard, Kevin                                     | 8 June 2026 | <a href="#">Open source</a> |
| 3 | <b>reStructuredText Primer - Sphinx Documentation</b> | Sphinx Documenta-<br>tion                           | 8 June 2026 | <a href="#">Open source</a> |
| 4 | <b>Welcome to the reStructuredText Primer!</b>        | reStructuredText<br>Primer 0.0.1 Docu-<br>mentation | 8 June 2026 | <a href="#">Open source</a> |

### 3.3.3 OpenTelemetry and Observability

| # | Source                                                                 | Creator                                                 | Date        | Link                        |
|---|------------------------------------------------------------------------|---------------------------------------------------------|-------------|-----------------------------|
| 5 | <b>How to Set Up a Complete Observability Stack with OpenTelemetry</b> | OneUptime   One<br>Complete Observ-<br>ability Platform | 6 Feb. 2026 | <a href="#">Open source</a> |
| 6 | <b>OpenTelemetry</b>                                                   | OpenTelemetry                                           | 8 June 2026 | <a href="#">Open source</a> |
| 7 | <b>open-telemetry/opentelemetry-<br/>proto</b>                         | OpenTelemetry -<br>CNCF                                 | 22 May 2019 | <a href="#">Open source</a> |
| 8 | <b>Python OpenTelemetry</b>                                            | OpenTelemetry                                           | 8 June 2026 | <a href="#">Open source</a> |
| 9 | <b>Prometheus Docs</b>                                                 | The Linux Founda-<br>tion                               | 9 June 2026 | <a href="#">Open source</a> |

### 3.3.4 Python Libraries and Tooling

| #  | Source                                                            | Creator                       | Date         | Link                        |
|----|-------------------------------------------------------------------|-------------------------------|--------------|-----------------------------|
| 10 | <b>SQLAlchemy</b>                                                 | SQLAlchemy                    | 8 June 2026  | <a href="#">Open source</a> |
| 11 | <b>SQLAlchemy ORM - SQLAlchemy 2.0 Documentation</b>              | SQLAlchemy Docu-<br>mentation | 8 June 2026  | <a href="#">Open source</a> |
| 12 | <b>Typer</b>                                                      | Typer                         | 8 June 2026  | <a href="#">Open source</a> |
| 13 | <b>SQLAlchemy Crash Course - Mas-<br/>ter Databases in Python</b> | Tim Ruscica                   | Mar 17, 2025 | <a href="#">Open source</a> |
| 14 | <b>socket — Low-level networking in-<br/>terface</b>              | Python Software<br>Foundation | Jun 10, 2026 | <a href="#">Open source</a> |
| 15 | <b>Python Requests post() Method</b>                              | W3Schools                     | 2026         | <a href="#">Open source</a> |

### 3.3.5 Software Testing and Code Coverage

| #  | Source                                                                                    | Creator                                                                | Date          | Link                        |
|----|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------|-----------------------------|
| 16 | <b>100% Test Coverage is a LIE, Here's Why...</b>                                         | Sam Newman                                                             | 8 June 2026   | <a href="#">Open source</a> |
| 17 | <b>Why 100% Test Coverage Isn't the Goal</b>                                              | Daniel Mauno Pettersson                                                | March 4, 2026 | <a href="#">Open source</a> |
| 18 | <b>Test Coverage</b>                                                                      | Fowler, Martin                                                         | 17 Apr. 2012  | <a href="#">Open source</a> |
| 19 | <b>Coverage Is Not Strongly Correlated with Test Suite Effectiveness</b>                  | Inozemtseva, Laura; Holmes, Reid                                       | 2014          | <a href="#">Open source</a> |
| 20 | <b>Code Coverage and Postrelease Defects: A Large-Scale Study on Open Source Projects</b> | Kochhar, Pavneet Singh; Lo, David; Lawall, Julia; Nagappan, Nachiappan | 2017          | <a href="#">Open source</a> |
| 21 | <b>A Brief Survey on Oracle-based Test Adequacy Metrics</b>                               | Hossain, Soneya Binta; Dwyer, Matthew B.                               | 2022          | <a href="#">Open source</a> |
| 22 | <b>Four Common Types of Code Coverage</b>                                                 | web.dev                                                                | 2024          | <a href="#">Open source</a> |
| 23 | <b>Getting 100% Code Coverage Doesn't Eliminate Bugs</b>                                  | Kapelonis, Kostis                                                      | 2017          | <a href="#">Open source</a> |

### 3.3.6 Logging

| #  | Source                                                       | Creator        | Date         | Link                        |
|----|--------------------------------------------------------------|----------------|--------------|-----------------------------|
| 24 | <b>Python Logging: Master the Module in Under 10 Minutes</b> | Tim van Cann   | Aug 22, 2024 | <a href="#">Open source</a> |
| 25 | <b>Python JSON Logger</b>                                    | Nicholas Hairs | 10 June 2026 | <a href="#">Open source</a> |
| 26 | <b>Logging in Python</b>                                     | GeeksforGeeks  | 10 June 2026 | <a href="#">Open source</a> |

### 3.3.7 Detailed Source Cards

#### 1. reStructuredText Markup Specification

**Type**

Specification

**Author**

Goodger, David

**Organization**

Docutils

**Date**

2026

**Accessed**

7 June 2026

**URL**

<https://docutils.sourceforge.io/docs/ref/rst/restructuredtext.html>

**i 2. How to Set Up a Complete Observability Stack with OpenTelemetry****Type**

Article

**Organization**

OneUptime | One Complete Observability Platform

**Date**

6 Feb. 2026

**Accessed**

7 June 2026

**URL**<https://oneuptime.com/blog/post/2026-02-06-complete-observability-stack-opentelemetry-open-source/view>**i 3. Restructuredtext Primer - Material for Sphinx****Type**

Documentation

**Author**

Sheppard, Kevin

**Accessed**

8 June 2026

**URL**<https://bashtage.github.io/sphinx-material/basics.html>**i 4. OpenTelemetry****Type**

Website

**Organization**

OpenTelemetry

**Accessed**

8 June 2026

**URL**<https://opentelemetry.io/>**i 5. open-telemetry/opentelemetry-proto****Type**

GitHub Repository

**Organization**

OpenTelemetry - CNCF

**Date**

22 May 2019

**Accessed**

8 June 2026

**URL**

<https://github.com/open-telemetry/opentelemetry-proto>

**Note**

Makefile referenced.

## 6. Python OpenTelemetry

**Type**

Documentation

**Organization**

OpenTelemetry

**Accessed**

8 June 2026

**URL**

<https://opentelemetry.io/docs/languages/python/>

## 7. reStructuredText Primer - Sphinx Documentation

**Type**

Documentation

**Organization**

Sphinx Documentation

**Accessed**

8 June 2026

**URL**

<https://www.sphinx-doc.org/en/master/usage/restructuredtext/basics.html>

## 8. SQLAlchemy

**Type**

Website

**Organization**

SQLAlchemy

**Accessed**

8 June 2026

**URL**

<https://www.sqlalchemy.org>



**i 9. SQLAlchemy ORM - SQLAlchemy 2.0 Documentation****Type**

Documentation

**Organization**

SQLAlchemy Documentation

**Accessed**

8 June 2026

**URL**<https://docs.sqlalchemy.org/en/20/orm/index.html>**i 10. Typer****Type**

Documentation

**Organization**

Typer

**Accessed**

8 June 2026

**URL**<https://typer.tiangolo.com/>**i 11. Welcome to the reStructuredText Primer!****Type**

Documentation

**Organization**

reStructuredText Primer 0.0.1 Documentation

**Accessed**

8 June 2026

**URL**<https://djdeangelis.github.io/reST-primer/index.html>**i 12. 100% Test Coverage is a LIE, Here's Why...****Type**

Video

**Author**

Sam Newman

**Organization**

Modern Software Engineering

**Accessed**

8 June 2026

**URL**

<https://www.youtube.com/watch?v=p1xZ-Ni2t8Q>

**i 13. SQLAlchemy Crash Course - Master Databases in Python**

**Type**

Video

**Author**

Tim Ruscica

**Organization**

NeuralNine

**Date**

Mar 17, 2025

**Accessed**

9 June 2026

**URL**

<https://www.youtube.com/watch?v=529LYDgRTgQ&t>

**i 14. Prometheus Docs**

**Type**

Documentation

**Organization**

The Linux Foundation

**Accessed**

9 June 2026

**URL**

<https://prometheus.io/docs/introduction/overview/>

**i 15. Why 100% Test Coverage Isn't the Goal**

**Author**

Daniel Mauno Pettersson

**Organization**

QA.tech

**Date**

March 4, 2026

**Accessed**

9 June 2026

**URL**

<https://qa.tech/blog/why-100-test-coverage-isnt-the-goal>

#### **i 16. Test Coverage**

**Type**

Article

**Author**

Fowler, Martin

**Organization**

Martin Fowler

**Date**

17 Apr. 2012

**Accessed**

9 June 2026

**URL**

<https://martinfowler.com/bliki/TestCoverage.html>

#### **i 17. Coverage Is Not Strongly Correlated with Test Suite Effectiveness**

**Type**

Conference Paper

**Author**

Inozemtseva, Laura; Holmes, Reid

**Organization**

ICSE 2014 / ACM

**Date**

2014

**Accessed**

9 June 2026

**URL**

<https://dl.acm.org/doi/10.1145/2568225.2568271>

#### **i 18. Code Coverage and Postrelease Defects: A Large-Scale Study on Open Source Projects**

**Type**

Journal Article

**Author**

Kochhar, Pavneet Singh; Lo, David; Lawall, Julia; Nagappan, Nachiappan

**Organization**

IEEE Transactions on Reliability

**Date**

2017  
**Accessed**  
9 June 2026  
**URL**  
[https://ink.library.smu.edu.sg/sis\\_research/3838/](https://ink.library.smu.edu.sg/sis_research/3838/)

#### 19. A Brief Survey on Oracle-based Test Adequacy Metrics

**Type**  
Research Survey  
**Author**  
Hossain, Soneya Binta; Dwyer, Matthew B.  
**Organization**  
arXiv  
**Date**  
2022  
**Accessed**  
9 June 2026  
**URL**  
<https://arxiv.org/abs/2212.06118>

#### 20. Four Common Types of Code Coverage

**Type**  
Technical Article  
**Organization**  
web.dev  
**Date**  
2024  
**Accessed**  
9 June 2026  
**URL**  
<https://web.dev/articles/ta-code-coverage>

#### 21. Getting 100% Code Coverage Doesn't Eliminate Bugs

**Type**  
Technical Article  
**Author**  
Kapelonis, Kostis  
**Organization**

Codepipes

**Date**

2017

**Accessed**

10 June 2026

**URL**

<https://blog.codepipes.com/testing/code-coverage.html>

**i 22. socket — Low-level networking interface****Type**

Documentation

**Organization**

Python Software Foundation

**Date**

Jun 10, 2026

**Accessed**

10 June 2026

**URL**

<https://docs.python.org/3/library/socket.html>

**i 23. Python Requests post() Method****Type**

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**Organization**

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10 June 2026

**URL**

[https://www.w3schools.com/PYTHON/ref\\_requests\\_post.asp](https://www.w3schools.com/PYTHON/ref_requests_post.asp)

**i 24. Python Logging: Master the Module in Under 10 Minutes****Type**

Video

**Author**

Tim van Cann

**Organization**

Timnology

**Date**

Aug 22, 2024

**Accessed**

10 June 2026

**URL**

<https://www.youtube.com/watch?v=A3FkYRN9qog>

**Note**

Was very helpful.

### 25. Python JSON Logger

**Type**

Documentation

**Author**

Nicholas Hairs

**Organization**

Python JSON Logger

**Accessed**

10 June 2026

**URL**

<https://nhairs.github.io/python-json-logger/>

### 26. Logging in Python

**Type**

Article

**Organization**

GeeksforGeeks

**Accessed**

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**URL**

<https://www.geeksforgeeks.org/python/logging-in-python/>

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## 3.4 Style Guide

This style guide is for me and serves as a reminder to me.

### 3.4.1 Scope of Guide

Seeing as I am the sole developer of this project, I don't think it is necessary to define all stylings. That being said, I will define some style choices.



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